

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 5: The Benchmark New
Keynesian Model



Prepared by Marcos Consuegra

What We Will Learn

- ▶ How to extend the basic monetary framework by introducing **nominal rigidities** through Calvo–Yun price setting.
- ▶ How price stickiness generates an **upward-sloping, forward-looking aggregate supply (NKPC)** and a non-neutral role for monetary policy.
- ▶ How intertemporal optimization by households leads to a **forward-looking aggregate demand (NK IS)** relation driven by the real interest rate.
- ▶ How to define and interpret the **natural rate of output** and distinguish it from the **efficient level** of output.
- ▶ How to combine the NKPC and the NK IS equation into a **log-linearized equilibrium system** describing short-run fluctuations.
- ▶ How monetary policy affects output and inflation through both the **policy rate** and **expectation management**.

Outline

- 1 Introduction
- 2 Model
- 3 Equilibrium with Flexible Prices
- 4 Price Rigidities
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- 6 Log-Linear Approximation
- 7 Main Takeaways

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The Effectiveness of Monetary Policy

- ▶ Friedman and Schwartz (1963) is a statement on the importance of **stabilization policy** by central banks.
- ▶ Three key insights:
 - ▶ **Natural rate of output:** long-run output is independent of monetary policy.
 - ▶ **Short-run monetary non-neutrality:** 'transient' price rigidities allow monetary shocks to influence real activity.
 - ▶ **Contractions and depressions:** major downturns are largely driven by contractions in the money supply.
- ▶ *"Prevention or moderation of the decline in the stock of money. . . would have reduced the contraction's severity and almost certainly its duration." (Friedman and Schwartz (1963), p. 301)*

Limitations of Real Business Cycle Theories

Real Business Cycle Models (Kydland and Prescott 1982)

- ▶ Fully microfounded: households and firms solve intertemporal optimization under resource constraints.
- ▶ Incorporate stochastic technology shocks and rational expectations.
- ▶ Imply **monetary neutrality**—policy cannot affect real variables.
- ▶ Cannot account for the empirically observed short-run effects of monetary policy.

Empirical Evidence: SVARs

- ▶ Monetary policy shocks stimulate output with a **lag of about six quarters**, with effects persisting for more than two years.
- ▶ These results provide empirical benchmarks for models of monetary transmission (Christiano, Eichenbaum, and Evans 1999; Woodford 2003).

Emergence of the New-Keynesian Theories

- ▶ Builds on RBC **microfoundations**: intertemporal optimization, rational expectations, and stochastic disturbances.
- ▶ Introduces **nominal rigidities** through imperfect competition and sticky prices (Mankiw 1985).
- ▶ Reconciles:
 - ▶ **Short-run monetary non-neutrality** (due to price stickiness).
 - ▶ **Long-run natural rate of output** (as in RBC).
- ▶ Provides a tractable framework consistent with SVAR evidence on the dynamic effects of monetary shocks.

Nominal Rigidities

- ▶ **Calvo–Yun price setting:** in each period, only a fraction $(1 - \alpha)$ of firms can reset their price, while the remaining fraction keeps an indexed price. This generates **staggered** price adjustments with random durations.
- ▶ Firms that reset understand that their price may remain fixed for several periods. As a result, they choose prices **anticipating future inflation and marginal costs**.
- ▶ This forward-looking behavior yields an **upward-sloping aggregate supply (NK Phillips Curve)**, where inflation depends on:
 - ▶ the current output gap, and
 - ▶ expected future inflation.
- ▶ Sticky prices create the mechanism for **short-run monetary non-neutrality**: by affecting real interest rates, monetary policy can influence output.

Connecting Theory and Evidence

- ▶ NKM bridges Friedman & Schwartz's empirical insights and RBC microfoundations.
- ▶ Captures realistic policy effects:
 - ▶ Output responds gradually to monetary shocks.
 - ▶ Inflation dynamics consistent with Calvo pricing.
- ▶ Provides a framework for evaluating optimal monetary policy (e.g., inflation targeting).

Limitations and Extensions

- ▶ Benchmark NKM: single sector, time-dependent Calvo pricing, homogeneous shocks.
- ▶ Micro evidence shows heterogeneity and state-dependent price adjustment (Nakamura and Steinsson 2008; Nakamura and Steinsson 2013).
- ▶ Calvo model remains a reasonable approximation for output response to a monetary shock.
- ▶ Extensions include banking sector, liquidity effects, and policy under financial crises (e.g., 2007–2008, COVID-19), which will be covered in later chapters.

Outline of the Chapter's Results

Flexible Prices: Classical Benchmark

- ▶ With fully flexible prices, the model reproduces the **classical dichotomy**.
- ▶ Aggregate supply is **vertical**: shifts in aggregate demand affect only the price level, not output.

Price Rigidities: New-Keynesian Benchmark

- ▶ Introducing **Calvo-style stickiness** yields an upward-sloping *AS* curve.
- ▶ Monetary policy becomes **non-neutral in the short run** and can stabilize output.

Forward-Looking Equations

- ▶ **Aggregate demand (NK IS)**: output depends negatively on the *current and expected future* real interest rate.
- ▶ **Aggregate supply (NKPC)**: inflation depends on the *present-discounted sum* of current and expected future output gaps.
- ▶ Monetary policy operates through the **policy rate** and by **shaping expectations**.

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Model Overview

Stochastic Environment

- ▶ Infinite-horizon, closed economy with uncertainty.
- ▶ The state of nature at time t is denoted by $s_t \in \mathcal{S}$, where the set $\mathcal{S}$ is invariant over time.
- ▶ The history up to time t is $s^t \equiv (s_t, s_{t-1}, \dots, s_{t_0})$.
- ▶ The probability distribution of histories is given by $f(s^t \mid s_{t_0})$ and has the Markov properties.

Agents

- ▶ Households
- ▶ Firms
- ▶ Government

Households: Preferences and Consumption

Expected Intertemporal Utility

$$E_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \xi_t \left[U(C_t) - \int_0^1 H(L_t(j)) dj \right] \right\}. \quad (5.1)$$

Interpretation

- ▶ $\beta \in (0, 1)$: subjective discount factor
- ▶ ξ_t : preference shock
- ▶ $U(\cdot)$: concave utility in a consumption bundle C
- ▶ $H(\cdot)$: convex disutility from hours of labor, supplied to firms producing good j .

Household: Consumption Bundle

Dixit–Stiglitz Aggregator

$$C_t = \left[\int_0^1 C_t(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}}, \quad \theta > 1. \quad (5.2)$$

Interpretation

- ▶ $C_t(j)$: consumption of variety $j \in [0, 1]$
- ▶ θ : elasticity of substitution across goods
- ▶ Higher $\theta \Rightarrow$ goods more substitutable
- ▶ $\theta \rightarrow \infty$: perfect substitutes $\Rightarrow C_t = \int_0^1 C_t(j) dj$

Household: Demand of Goods

Demand for Variety j

$$C_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} C_t. \quad (5.3)$$

- Demand depends on the relative price of good j , $P_t(j)$, with respect to the aggregate price index P_t :

Aggregate Price Index

$$P_t = \left[\int_0^1 P_t(j)^{1-\theta} dj \right]^{\frac{1}{1-\theta}}. \quad (5.4)$$

Derivation of demand and price index in Appendix

Households: Consumption, Saving, and Labor Supply

Optimization Problem

- ▶ At each point in time and contingencies, households choose consumption, saving, and hours worked.
- ▶ They plan savings using available assets to smooth consumption across future states.
- ▶ Financial markets are assumed to be **complete**, allowing the households to implement arbitrary patterns of portfolio payoffs across successor states.

Complete Financial Markets

Complete Markets

- ▶ Denote $\mathcal{W}_t(s_{t+1})$ as the desired random payoff in state s_{t+1} at $t + 1$, conditional on history s^t .
- ▶ Complete markets mean any desired random payoff $\mathcal{W}_t(s_{t+1})$ can be replicated by portfolios.
- ▶ Number of traded assets $\geq$ number of possible states at $t + 1$.
- ▶ No-arbitrage: portfolios with identical payoffs have identical prices.

Complete Financial Markets

Complete markets can be characterized by a set of state-contingent securities spanning all states of nature:

- ▶ A state-contingent security, at time t , pays 1 unit of currency in a specific state s_{t+1} at date $t + 1$, and 0 otherwise.
- ▶ Price of the security at t : $\frac{v_t(s_{t+1})}{1 + i_t}$, conditional on history s^t .
- ▶ Note: Normalization by $(1 + i_t)$ is for analytical convenience; i_t is the nominal risk-free rate.

Portfolio Valuation under Complete Markets

- ▶ A generic portfolio with random payoff $\mathcal{W}_{t+1}$ has value:

$$B_t = \sum_{s_{t+1} \in S} \frac{v_t(s_{t+1})}{1 + i_t} \mathcal{W}_t(s_{t+1}).$$

- ▶ Each payoff $\mathcal{W}_{t+1}$ is valued through the price of the respective state-contingent security, $\frac{v_t(s_{t+1})}{1+i_t}$.
- ▶ No-arbitrage implies identical prices for portfolios with identical payoffs.

Stochastic Discount Factor (SDF)

$$\tilde{R}_t(s_{t+1}) \equiv \frac{v_t(s_{t+1})}{f_t(s_{t+1})(1 + i_t)},$$

where $f_t(s_{t+1}) \equiv f_t(s_{t+1}|s^t)$.

Portfolio Valuation under Complete Markets

- Using the stochastic discount factor into the valuation of the portfolio:

$$\begin{aligned} B_t &= \sum_{s_{t+1} \in S} f_t(s_{t+1}) \tilde{R}_t(s_{t+1}) \mathcal{W}_t(s_{t+1}) \\ &= E_t \tilde{R}_{t,t+1} \mathcal{W}_{t+1}. \end{aligned} \tag{5.5}$$

Interpretation

- The price of a portfolio with a random return $\mathcal{W}_{t+1}$ is equal to the expected discounted value of its payoff, where the discount factor is represented by the random variable $\tilde{R}_{t,t+1} \equiv \tilde{R}_t(s_{t+1})$.
- Note: For simplicity, $\mathcal{W}_{t+1}$ is used in place of $\mathcal{W}_{t,t+1}$.

Risk-Free Bond and the Stochastic Discount Factor

Risk-Free Portfolio

- ▶ Consider a risk-free portfolio paying $\mathcal{W}_t(s_{t+1}) = 1$ for each $s_{t+1} \in \mathcal{S}$, the portfolio replicates the price of a risk-free nominal bond $B_t = \frac{1}{1+i_t}$.
- ▶ Using (5.5):

$$\frac{1}{1+i_t} = E_t[\tilde{R}_{t,t+1}],$$

implies for state-contingent prices:

$$\sum_{s_{t+1} \in \mathcal{S}} v_t(s_{t+1}) = 1.$$

- ▶ The sum of state-contingent security prices equals one, consistent with no-arbitrage.

Household: Flow Budget Constraint

Budget Constraint at time t

$$\underbrace{B_t + P_t C_t}_{\text{total spending}} = \underbrace{\mathcal{W}_t + \int_0^1 W_t(j) L_t(j) dj + \Psi_t - T_t}_{\text{total resources net of taxes}} \quad (5.6)$$

Interpretation

- ▶ B_t : value of the portfolio with state-contingent payoffs $\mathcal{W}_{t+1}$.
- ▶ $P_t C_t$: nominal consumption spending .
- ▶ $\mathcal{W}_t$: payoff from past portfolio holdings.
- ▶ $\int_0^1 W_t(j) L_t(j) dj$: income from each variety of labor j supplied.
- ▶ Ψ_t : firms' profits
- ▶ T_t : Taxes.
- ▶ Note: Cash is neglected, since it is only held at $i_t = 0$, imposing a ZLB.

Household: Borrowing Limit Condition

Borrowing Limit:

$$-\mathcal{W}_t \leq E_t \left\{ \sum_{T=t}^{\infty} \tilde{R}_{t,T} (W_T L_T + \Psi_T - T_T) \right\} < \infty,$$

for each time $t \geq t_0$ and contingency at time t ; **or**, alternatively:

Intertemporal Budget Constraint:

$$\underbrace{E_t \left\{ \sum_{T=t}^{\infty} \tilde{R}_{t,T} (P_T C_T) \right\}}_{\text{Expected PDV of Consumption}} \leq \underbrace{\mathcal{W}_t + E_t \left\{ \sum_{T=t}^{\infty} \tilde{R}_{t,T} (W_T L_T + \Psi_T - T_T) \right\}}_{\text{Portfolio Payoff and PDV of Net Income}}. \quad (5.7)$$

Representative Consumer: Optimization Problem

- ▶ The representative consumer chooses stochastic sequences $\{C_t, \mathcal{W}_{t+1}, L_t\}_{t=t_0}^{\infty}$ to maximize expected intertemporal utility:

$$E_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \xi_t \left[U(C_t) - \int_0^1 H(L_t(j)) dj \right] \right\}.$$

Subject to:

- ▶ Portfolio valuation (5.5)
- ▶ Budget constraint (5.6)
- ▶ Intertemporal budget constraint (5.7)
- ▶ Non-negativity constraints $C_t, L_t \geq 0$
- ▶ Given initial condition $\mathcal{W}_{t_0}$

Household Optimality: Euler Equation

- ▶ The following first-order condition characterizes the representative household's optimal consumption/saving choice at time t :

Euler Equation

$$\frac{\xi_t U_c(C_t)}{P_t} = \frac{\beta}{\tilde{R}_{t,t+1}} \frac{\xi_{t+1} U_c(C_{t+1})}{P_{t+1}}. \quad (5.8)$$

- ▶ $U_c(\cdot)$ is the marginal utility of consumption.
- ▶ For each state at $t + 1$, there is a corresponding Euler equation.
- ▶ Complete markets allow households to allocate consumption optimally across all contingencies at $t + 1$.

Household Optimality: Euler Equation

- The Euler equations can be expressed in terms of state-contingent securities as:

$$\underbrace{\frac{v_t(s_{t+1})}{1+i_t} \frac{\xi_t U_c(C_t)}{P_t}}_{\text{marginal cost of saving one unit today, delivering one unit of currency in state } s_{t+1}} = \underbrace{\beta f_t(s_{t+1}) \frac{\xi_{t+1} U_c(C_{t+1})}{P_{t+1}}}_{\text{discounted marginal benefit of consuming one unit tomorrow in state } s_{t+1}} . \quad (5.9)$$

- The benefit is positive with probability $f_t(s_{t+1})$.

Household Optimality: Stochastic Euler Equation

- ▶ Multiplying (5.8) by $\tilde{R}_{t,t+1}$ and taking expectations at time t yields:

Stochastic Euler Equation

$$\xi_t U_c(C_t) = \beta(1 + i_t) E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(C_{t+1}) \right\}. \quad (5.10)$$

- ▶ Having used $E_t \tilde{R}_{t,t+1} = 1/(1 + i_t)$.
- ▶ **Note:** (5.10) is implied by the full set of state-contingent Euler equations (5.8), but it does *not* imply them.

Household Optimality: Labor Supply

Optimal Labor Supply for each variety of labor $j \in [0, 1]$

$$\underbrace{\frac{H_l(L_t(j))}{U_c(C_t)}}_{\text{Marginal Rate of Substitution}} = \underbrace{\frac{W_t(j)}{P_t}}_{\text{Real Wage for variety } j} \quad (5.11)$$

- ▶ $H_l(\cdot)$ denotes the derivative of disutility of labor with respect to hours supplied.
- ▶ Concavity of $U(\cdot)$ and convexity of $H(\cdot)$ imply labor supply is increasing in the real wage and decreasing in consumption.
- ▶ At the optimum, the intertemporal budget constraint (5.7) holds with equality at all times and in all states.

Firms: Market Structure and Pricing Power

- ▶ Firms operate in a market characterized by **monopolistic competition**: goods are differentiated, not perfect substitutes.
- ▶ Each firm can influence its demand by setting $P_t(j)$ through the demand function (5.12).
- ▶ Firms are *small* relative to the overall market, taking aggregate price P_t and quantities (C_t, G_t) as given.

Firms: Demand and Technology

- ▶ Aggregate demand for firm j depends on both private demand $C_t(j)$ and government demand $G_t(j)$:

Demand for Firm j 's Output

$$Y_t(j) = C_t(j) + G_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} (C_t + G_t). \quad (5.12)$$

- ▶ Assuming similar demand of G , i.e., $G_t(j) = (P_t(j)/P_t)^{-\theta} G_t$.

Production Technology

$$Y_t(j) = A_t L_t(j) \quad (5.13)$$

- ▶ A_t : stochastic labor productivity shock, identical across firms.

Firms: Profits

Profit Function for Firm j

$$\begin{aligned}\Psi_t(j) &= (1 + \tau_t^s)P_t(j)Y_t(j) - W_t(j)L_t(j) \\ &= \left[(1 + \tau_t^s)P_t(j) - \frac{W_t(j)}{A_t} \right] Y_t(j),\end{aligned}\tag{5.14}$$

- ▶ τ_t^s : stochastic revenue subsidy.
- ▶ Note: The subsidy is useful to introduce a stochastic markup shifter and ensure an efficient steady-state equilibrium (simplifying the analysis of optimal policy in Chapter 7).

Government: Consolidated Budget Constraint

Government Budget Constraint

$$\begin{aligned} B_t^g &= (1 + i_{t-1})B_{t-1}^g + \int_0^1 P_t(j)G_t(j) dj + \tau_t^s \int_0^1 P_t(j)Y_t(j) dj - T_t \\ &= (1 + i_{t-1})B_{t-1}^g + P_t G_t + \tau_t^s P_t Y_t - T_t. \end{aligned} \quad (5.15)$$

- ▶ B_t^g : government debt; T_t : lump-sum taxes.
- ▶ $P_t G_t$: nominal government spending; $\tau_t^s P_t Y_t$: subsidy to all firms.
- ▶ Second line uses $G_t(j) = (P_t(j)/P_t)^{-\theta} G_t$, $Y_t(j) = (P_t(j)/P_t)^{-\theta} Y_t$, and the aggregate price index (5.4).
- ▶ The central bank backs treasury debt, i.e., government debt is unconstrained by solvency conditions (see Chapter 1).

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Equilibrium: Flexible Prices

- ▶ With flexible prices, each firm j sets $P_t(j)$ to maximize profits (5.14) given demand (5.12):

Optimal Pricing Rule

$$P_t(j) = \underbrace{\mu_t \frac{W_t(j)}{A_t}}_{\text{Markup over nominal marginal cost}}, \quad \mu_t \equiv \frac{1}{1 + \tau_t^s} \frac{\theta}{\theta - 1}$$

- ▶ Rewriting, we note that firm j 's labor demand is flat at the real wage:

Labor Demand

$$\frac{W_t(j)}{P_t(j)} = \frac{A_t}{\mu_t}, \quad \text{constant across all } j.$$

Derivation of Optimal Pricing Rule in Appendix

Equilibrium: Labor and Prices

- ▶ Household labor supply (5.11) combined with firm labor demand yields:

$$\frac{H_l(L_t(j))}{U_c(C_t)} = \frac{A_t}{\mu_t} \frac{P_t(j)}{P_t}, \quad \forall j$$

- ▶ Convexity of $H(\cdot)$ implies a symmetric equilibrium:
 - ▶ Labor is identical across firms: $L_t(j) = L_t$.
 - ▶ Output is equal across goods: $Y_t(j) = Y_t$.
 - ▶ Prices are uniform across goods: $P_t(j) = P_t$.

Equilibrium: Natural Rate of Output

- ▶ Substituting $L_t = Y_t/A_t$ and using goods market equilibrium $C_t = Y_t - G_t$ gives:

Natural Level of Output Y_t^n

$$\frac{H_l\left(\frac{Y_t^n}{A_t}\right)}{U_c(Y_t^n - G_t)} = \frac{A_t}{\mu_t}. \quad (5.16)$$

Properties:

- ▶ $Y_t^n \uparrow$ in A_t and G_t ; $Y_t^n \downarrow$ in μ_t .
- ▶ Higher productivity $\rightarrow$ firms hire more labor $\rightarrow$ output rises.
- ▶ Higher government spending $\rightarrow$ positive wealth effect $\rightarrow$ labor supply and output rise, but consumption falls.
- ▶ Higher markup $\rightarrow$ labor demand falls $\rightarrow$ output falls.

Flexible-Price Equilibrium

- ▶ **Classical Dichotomy:** Equation (5.16) implies that in the flexible-price equilibrium, the real allocation is independent of monetary policy.
- ▶ Monetary policy **does** influence nominal variables such as interest rates and prices.
- ▶ The analysis from Chapter 1 applies, with the caveat that the economy is stochastic and output is now endogenously determined.

Efficient vs. Natural Output

- ▶ Assume the central planner chooses Y_t to maximize consumer utility (5.1) subject to the resource constraint $Y_t = C_t + G_t$ and production technology $Y_t = A_t L_t$ implies:

Efficient Level of Output Y_t^e

$$\frac{H_\ell\left(\frac{Y_t^e}{A_t}\right)}{U_c(Y_t^e - G_t)} = A_t. \quad (5.17)$$

- ▶ Comparing (5.16) and (5.17):
 - ▶ **Markup shock** $\rightarrow$ **inefficient shock** (affect Y_t^n but not Y_t^e).
 - ▶ **Productivity & government spending shock** $\rightarrow$ **efficient shocks** (affect Y_t^n and Y_t^e in the same proportion).

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Price Rigidities: Calvo Pricing

- ▶ **Monetary policy neutrality** can be broken by introducing **price rigidities**.
- ▶ Benchmark mechanism: **Calvo (1983) / Yun (1996)**.

Setup

- ▶ Continuum of firms $j \in [0, 1]$; each period, a fraction $(1 - \alpha)$ can reset their price.
- ▶ A price set at time t remains for next period $t + 1$ with probability α , and for period $T > t$ with probability α^{T-t} .
- ▶ **Non-reset prices** are indexed to the inflation target Π : if $P_t(j)$ persists until T , then $P_T(j) = P_t(j)\Pi^{T-t}$.
- ▶ Adjusting firms choose prices to maximize the **expected discounted value of profits**, taking into account the probability of the price remaining —appropriately indexed.

Price Rigidities: Profits

Profits at $T > t$ conditional on the price chosen at time t

$$\begin{aligned}\Psi_T(j) &= (1 + \tau_T^s)P_t(j)\Pi^{T-t}Y_T(j) - W_T(j)L_T(j) \\ &= (1 + \tau_T^s)P_t(j)\Pi^{T-t}Y_T(j) - \frac{W_T(j)}{A_T}Y_T(j)\end{aligned}$$

Having used the production function: $L_T(j) = Y_T(j)/A_T$.

Demand at time T conditional on price set at t :

$$Y_T(j) = \left(\frac{P_t(j)\Pi^{T-t}}{P_T} \right)^{-\theta} Y_T \quad (5.18)$$

Price Rigidities: Optimization Problem

Expected present-discounted value of profits for a firm j that has set prices at time t

$$E_t \sum_{T=t}^{\infty} \alpha^{T-t} \tilde{R}_{t,T} \left[(1 + \tau_T^s) \Pi^{T-t} P_t(j) Y_T(j) - \frac{W_T(j)}{A_T} Y_T(j) \right]. \quad (5.19)$$

- ▶ Firm j chooses $P_t(j)$ to maximize (5.19) given demand (5.18).
- ▶ **Note:** Wages $W_T(j)$ are taken as given (firms are small in their sector, wage-takers).

Price Rigidities: Optimality Condition

First-order condition for optimal price $P_t(j)$:

$$\frac{P_t(j)}{P_t} = \frac{E_t \left\{ \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \xi_T U_c(C_T) \left(\frac{P_T}{P_t} \frac{1}{\Pi^{T-t}} \right)^{\theta} \frac{W_T(j)}{A_T P_T} Y_T \right\}}{E_t \left\{ \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \xi_T U_c(C_T) \left(\frac{P_T}{P_t} \frac{1}{\Pi^{T-t}} \right)^{\theta-1} \frac{Y_T}{\mu_T} \right\}}. \quad (5.20)$$

- ▶ Discount factor: $\tilde{R}_{t,T} = \beta^{T-t} \frac{\xi_T U_c(C_T)}{\xi_t U_c(C_t)} \frac{P_t}{P_T}$ using (5.8).
- ▶ Recall, $\mu_t = \theta / [(1 + \tau_t^s)(\theta - 1)]$.
- ▶ In equilibrium, all adjusting firms set $P_t(j) = \mathcal{P}_t^*$.
- ▶ Uniqueness is ensured because the LHS increases in $P_t(j)$ and the RHS decreases due to dependence of $W_T(j)$ on $P_t(j)$.

Price Rigidities: Optimality Condition

- Assume isoelastic disutility of labor:

$$H(L(j)) = \frac{L(j)^{1+\eta}}{1+\eta}, \quad \eta \geq 0 \text{ (Frisch elasticity).}$$

- Using (5.11), $L_T(j) = Y_T(j)/A_T$ and (5.18), we can substitute $W_T(j)/P_T$ in (5.20) to simplify the optimal price condition:

Optimal Price with Isoelastic Labor Disutility

$$\left(\frac{\mathcal{P}_t^*}{P_t}\right)^{1+\theta\eta} = \frac{E_t \left\{ \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \xi_T \left(\frac{P_T}{P_t} \frac{1}{\Pi^{T-t}}\right)^{\theta(1+\eta)} \left(\frac{Y_T}{A_T}\right)^{1+\eta} \right\}}{E_t \left\{ \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \xi_T U_c(C_T) \left(\frac{P_T}{P_t} \frac{1}{\Pi^{T-t}}\right)^{\theta-1} \frac{Y_T}{\mu_T} \right\}}. \quad (5.21)$$

Price Rigidities: General Price Index

- ▶ Remaining fraction α of firms (not adjusting) index their previous prices to the inflation target Π .
- ▶ Using the law of large numbers, recursive aggregation of all past price vintages implies:

Law of Motion for the Aggregate Price Index:

$$P_t^{1-\theta} = (1 - \alpha)(\mathcal{P}_t^*)^{1-\theta} + \alpha \Pi^{1-\theta} P_{t-1}^{1-\theta}, \quad (5.22)$$

Derivation of Price Index Law of Motion

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New Keynesian Aggregate Supply (AS)

- Substituting into (5.21) the goods market equilibrium $Y_t = C_t + G_t$, labor supply (5.11), and using (5.22) for $\mathcal{P}_t^*/P_t$, we obtain:

NK Aggregate Supply (AS):

$$\left(\frac{1 - \alpha \left(\frac{\Pi_t}{\Pi} \right)^{\theta-1}}{1 - \alpha} \right)^{\frac{1+\theta\eta}{\theta-1}} = \frac{F_t}{J_t}, \quad (5.23)$$

with

$$F_t = \xi_t U_c(Y_t - G_t) \frac{Y_t}{\mu_t} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta-1} F_{t+1} \right\}, \quad (5.24)$$

$$J_t = \xi_t \left(\frac{Y_t}{A_t} \right)^{1+\eta} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta(1+\eta)} J_{t+1} \right\}. \quad (5.25)$$

Derivation in Appendix

New Keynesian Aggregate Demand (AD)

- Using the goods market equilibrium condition $Y_t = C_t + G_t$, the stochastic Euler equation (5.10) can be rewritten as:

NK Aggregate Demand (AD)

$$\xi_t U_c(Y_t - G_t) = \beta(1 + i_t)E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(Y_{t+1} - G_{t+1}) \right\} \quad (5.26)$$

Intertemporal Resource Constraint of the Economy

Intertemporal Resource Constraint of the Economy

$$\frac{B_{t-1}^g}{P_t} = E_t \left\{ \sum_{T=t}^{\infty} \beta^{T-t} \frac{\xi_T U_c(Y_T - G_T)}{\xi_t U_c(Y_t - G_t)} \left(\frac{T_T}{P_T} - \tau_T^s Y_T - G_T \right) \right\} \quad (5.27)$$

- (5.27) is a mirror image of the intertemporal budget constraint of the households through goods and asset markets equilibrium.

Equilibrium with Price Rigidities

Equilibrium: A set of stochastic processes

$$\{P_t, i_t, Y_t, F_t, J_t, T_t, B_t^g\}_{t=t_0}^{\infty}$$

that satisfies:

- ▶ Aggregate demand equation (5.26),
- ▶ Aggregate supply block (5.23), (5.24), (5.25),
- ▶ Government flow budget constraint (5.15),
- ▶ Intertemporal resource constraint (5.27),

given the zero lower bound $i_t \geq 0$, exogenous stochastic processes $\{\xi_t, A_t, G_t, \tau_t^s\}_{t=t_0}^{\infty}$, and appropriate initial conditions.

- ▶ The two remaining degrees of freedom are exhausted by specifying monetary and fiscal policy through the stochastic processes $\{T_t, i_t\}_{t=t_0}^{\infty}$.

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Log-Linear Approximation: Motivation

- ▶ Closed-form solutions for the full nonlinear equilibrium are infeasible.
- ▶ Alternative: approximate equilibrium dynamics around the steady state.
- ▶ Two main steps:
 - ① Characterize the steady-state equilibrium.
 - ② Linearize the equilibrium conditions around the steady state.
- ▶ This allows analyzing how small shocks perturb the economy around its steady-state values.

Steady-State Assumptions

Notation: Steady-state values are time-invariant and written without time subscripts.

- All exogenous processes constant in steady state:

$$\xi_t = \xi, \quad A_t = A, \quad G_t = G, \quad \mu_t = 1.$$

- Tax-subsidy set to remove steady-state mark-up distortion:

$$\tau^s = \frac{1}{\theta - 1}.$$

- Monetary policy pins down inflation at its target level:

$$\Pi_t = \Pi.$$

Steady-State Relations

- In steady state, the equilibrium relations simplify as follows:

Steady-State Relations:

$$1 + i = \beta^{-1}\Pi, \quad \text{from (5.26)} \quad (5.28)$$

$$F = J, \quad \text{from (5.23)} \quad (5.29)$$

$$F = \frac{\xi U_c(Y - G)Y}{1 - \alpha\beta}, \quad \text{from (5.24)} \quad (5.30)$$

$$J = \frac{\xi(Y/A)^{1+\eta}}{1 - \alpha\beta}, \quad \text{from (5.25)} \quad (5.31)$$

$$(Y/A)^\eta = A U_c(Y - G), \quad \text{from preceding equations} \quad (5.32)$$

Implication: Steady-state output is the efficient output level, see (5.17).

Log-Linearization Notation

Notation:

- ▶ $\hat{X}_t \equiv \ln(X_t/X)$: log deviation of variable X from its steady state.
- ▶ $\hat{i}_t = \ln[(1 + i_t)/(1 + i)]$, $\pi_t = \ln \Pi_t$, $\pi = \ln \Pi$.
- ▶ $\hat{G}_t = (G_t - G)/Y$, $\hat{\xi}_t = \ln(\xi_t/\xi)$, $\hat{\mu}_t = \ln(\mu_t/\mu)$.
- ▶ Intertemporal elasticity of substitution: $\sigma \equiv -U_c/(U_{cc}Y)$, evaluated at steady state.

Aggregate Supply (NK Phillips Curve)

New Keynesian Phillips Curve:

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi), \quad (5.33)$$

Slope of the AS curve:

$$\kappa \equiv \frac{1 - \alpha}{\alpha} \frac{(1 - \alpha\beta)(\sigma^{-1} + \eta)}{1 + \theta\eta}. \quad (5.34)$$

Properties:

- ▶ κ increases when prices are less sticky ($\alpha \downarrow$).
- ▶ Lower θ (weaker strategic complementarity) $\Rightarrow$ steeper AS.
- ▶ Lower η (more elastic labor supply) $\Rightarrow$ higher κ .

Aggregate Demand (NK IS Equation)

New Keynesian IS Equation:

$$\hat{Y}_t = \hat{G}_t + E_t(\hat{Y}_{t+1} - \hat{G}_{t+1}) - \sigma \left[\underbrace{\hat{i}_t - E_t(\pi_{t+1} - \pi)}_{\text{real interest rate}} + E_t(\hat{\xi}_{t+1} - \hat{\xi}_t) \right] \quad (5.35)$$

Properties:

- ▶ Output depends negatively on the **real interest rate**.
- ▶ Fiscal demand $\hat{G}_t$ and preference shocks $\hat{\xi}_t$ act as **direct shifters** of aggregate demand.

Natural vs. Efficient Output

Natural Output:

$$\hat{Y}_t^n = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t - \frac{1}{\sigma^{-1} + \eta} \hat{\mu}_t. \quad (5.36)$$

Efficient Output:

$$\hat{Y}_t^e = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t. \quad (5.37)$$

- ▶ Markup shocks reduce Y_t^n but leave Y_t^e unchanged (inefficient shock).
- ▶ Productivity and public spending raise both Y_t^n and Y_t^e .
- ▶ Public spending multiplier < 1 unless labor disutility is linear ($\eta = 0$).
- ▶ Productivity multiplier > 1 if $\sigma > 1$; < 1 if $\sigma < 1$.

Derivation of Log-Linearized System

Log-Linearized NK Model

- ▶ The log-linearized AD and AS schedules form a system of two equilibrium conditions in two unknowns: output $\hat{Y}_t$ and inflation π_t , given a monetary policy path for the nominal interest rate.
- ▶ Unlike the models in Chapters 1–4 where monetary policy was neutral on output, the presence of price rigidities allows monetary policy to affect output.
- ▶ **Note:** Fiscal policy is considered as *passive* in local analysis so that constraints (5.15) and (5.27) do not restrict equilibrium. Global determinacy, however, requires *active* fiscal policy to rule out multiple equilibria (see Chapter 1).

Aggregate Supply: Key Implications

New Keynesian Phillips Curve:

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi).$$

- ▶ With sticky prices, the AS curve is **upward sloping**: inflation increases when output exceeds its natural level.
- ▶ The **output gap** ($\hat{Y}_t - \hat{Y}_t^n$) is the key driver of inflation dynamics.
- ▶ Higher output raises marginal costs, which pushes inflation upward.

Aggregate Supply: Key Implications

- **Forward-looking feature:** expected future output gaps influence current inflation.

$$\pi_t - \pi = \kappa E_t \sum_{T=t}^{\infty} \beta^{T-t} (\hat{Y}_T - \hat{Y}_T^n).$$

- This captures the fact that firms set prices today anticipating future marginal costs and inflation.
- **Neutrality benchmark:** If all firms reoptimize prices ($\alpha = 0$), then $\kappa \rightarrow \infty$, the AS curve becomes vertical, and $\hat{Y}_t = \hat{Y}_t^n$ in every period.

Aggregate Demand: Key Implications

New Keynesian IS Equation:

$$\hat{Y}_t = \hat{G}_t + E_t(\hat{Y}_{t+1} - \hat{G}_{t+1}) - \sigma [\hat{i}_t - E_t(\pi_{t+1} - \pi) + E_t(\hat{\xi}_{t+1} - \hat{\xi}_t)] .$$

- ▶ The *AD* equation features:
 - ① A **negative relation** between output and the real interest rate.
 - ② A **forward-looking** component.
- ▶ **Real interest rate channel:** A higher real rate,

$$\hat{i}_t - E_t(\pi_{t+1} - \pi),$$

raises saving, lowers consumption, and reduces output. Sensitivity increases with the intertemporal elasticity (σ).

Aggregate Demand: Key Implications

► **Monetary policy transmission:**

- ① Adjusting the nominal rate $\hat{i}_t$,
- ② Influencing expected inflation $E_t(\pi_{t+1} - \pi)$.

► **Forward-looking feature:** Iterating the AD equation (5.35) forward:

$$\hat{Y}_t = \hat{G}_t + \sigma \hat{\xi}_t - \sigma E_t \sum_{T=t}^{\infty} (\hat{i}_T - (\pi_{T+1} - \pi)) + \lim_{T \rightarrow \infty} (\hat{Y}_T - \hat{G}_T - \sigma \hat{\xi}_T).$$

- Current output reflects the **entire expected path of real interest rates**. Expectations of future policy shifts influence demand immediately.

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Main Takeaways

- ▶ **Flexible prices:** the model reproduces the **classical dichotomy**; output is pinned down by real fundamentals ($Y_t = Y_t^n$).
- ▶ **Sticky prices (Calvo–Yun):** nominal rigidities generate a **forward-looking NK Phillips Curve**: inflation depends on the **output gap** and expected future inflation.
- ▶ **Intertemporal demand:** the **NK IS equation** links output to the **expected path of real interest rates**; current demand reacts to both current and anticipated future policy.
- ▶ **Natural vs. efficient output:** markup shocks move Y_t^n but not Y_t^e (inefficient shocks), while productivity and spending shocks move both.
- ▶ **Policy lesson:** monetary policy affects real activity in the short run through the policy rate and **expectation management**; the benchmark NK system is the foundation for the AS–AD analysis in Chapter 6.

8 Appendix: Derivations

9 Bibliography

Derivation of equations (5.3) and (5.4)

The price index (P) is defined as the minimum expenditure needed to purchase one unit of consumption good (C). The total expenditure is represented by

$$\mathcal{T}_t = \int_0^1 P_t(j) C_t(j) dj,$$

and P_t is such that

$$P_t = \min_{\{C_t(j)\}} \mathcal{T}_t$$

subject to

$$C_t = \left[\int_0^1 C_t(j)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{\theta-1}} = 1.$$

The Lagrangian of the problem is represented by

$$\mathcal{L} = \int_0^1 P_t(j)C_t(j)dj - \lambda \left[\left(\int_0^1 C_t(j)^{\frac{\theta-1}{\theta}} dj \right)^{\frac{\theta}{\theta-1}} - 1 \right]$$

whose first-order conditions are

$$\begin{aligned} P_t(j) &= \lambda \left(\int_0^1 C_t(j)^{\frac{\theta-1}{\theta}} dj \right)^{\frac{\theta}{\theta-1}-1} C_t(j)^{-\frac{1}{\theta}} \\ &= \lambda C_t^{\frac{1}{\theta}} C_t(j)^{-\frac{1}{\theta}} \end{aligned} \tag{5.38}$$

for each j . Inserting this result into the expenditure function, we obtain

$$\mathcal{T}_t = \int_0^1 P_t(j)C_t(j)dj = \lambda C_t^{\frac{1}{\theta}} \int_0^1 C_t(j)^{1-\frac{1}{\theta}} dj = \lambda C_t.$$

Therefore, given that P_t is defined as the minimum expenditure $\mathcal{T}_t$ to purchase one unit of C_t , then setting $P_t = \mathcal{T}_t$ and $C_t = 1$ in the above equation, it follows that $P_t = \lambda$. Taking $C_t(j)$ from (5.38) and inserting it into the expenditure function and using $P_t = \lambda$ and $C_t = 1$, it follows that

$$P_t = \left(\int_0^1 P_t(j)^{1-\theta} dj \right)^{\frac{1}{1-\theta}}$$

which is (5.4). Demand (5.3) can be obtained from (5.38) using $P_t = \lambda$.

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Derivation of Law of Motion of Price Index (5.22)

The expression (5.22) can be derived formally by noting that

$$\begin{aligned} P_t^{1-\theta} &= (1-\alpha) \sum_{j=0}^{\infty} \alpha^j (\Pi^j \mathcal{P}_{t-j}^*)^{1-\theta} \\ &= (1-\alpha)(\mathcal{P}_t^*)^{1-\theta} + \alpha \Pi^{1-\theta} \left\{ (1-\alpha) \sum_{j=0}^{\infty} \alpha^j (\Pi^j \mathcal{P}_{t-1-j}^*)^{1-\theta} \right\} \\ &= (1-\alpha)(\mathcal{P}_t^*)^{1-\theta} + \alpha \Pi^{1-\theta} P_{t-1}^{1-\theta}, \end{aligned}$$

having used the law of large number. The first line says that at time t a mass $(1-\alpha)$ sets the price $\mathcal{P}_t^*$, a mass $(1-\alpha)\alpha$ sets $\mathcal{P}_{t-1}^* \Pi$, a mass $(1-\alpha)\alpha^2$ sets $\mathcal{P}_{t-2}^* \Pi^2$ and so forth. From the second to the third line, the term in the curly bracket is $P_{t-1}^{1-\theta}$.

[Back to Calvo Price Index](#)

Flexible-price markup rule: derivation sketch

Maximize $\Psi_t(j) = (1 + \tau_t^s) P_t(j) Y_t(j) - \frac{W_t(j)}{A_t} Y_t(j)$ s.t. $Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\theta} (C_t + G_t)$.

FOC wrt $P_t(j)$:

$$\frac{\partial \Psi_t(j)}{\partial P_t(j)} = (1 + \tau_t^s) Y_t(j) + (1 + \tau_t^s) P_t(j) \frac{\partial Y_t(j)}{\partial P_t(j)} - \frac{W_t(j)}{A_t} \frac{\partial Y_t(j)}{\partial P_t(j)} = 0,$$

$$\frac{\partial Y_t(j)}{\partial P_t(j)} = -\theta \left(\frac{P_t(j)}{P_t} \right)^{-\theta-1} \frac{1}{P_t} (C_t + G_t) = -\frac{\theta}{P_t(j)} Y_t(j).$$

Substitute and simplify $\Rightarrow P_t(j) = \frac{\theta}{(1+\tau_t^s)(\theta-1)} \frac{W_t(j)}{A_t} = \mu_t \frac{W_t(j)}{A_t}$.

[Back to Optimal Price](#)

Derivation of Aggregate Supply

Start from the optimal reset price expression and use labor-supply

$W_T/P_T = H_\ell(L_T)/U_c(C_T)$ with $L_T = Y_T(j)/A_T$ and demand. After algebra and using the price-index law to replace $\mathcal{P}_t^*/P_t$, define

$$F_t \equiv \xi_t U_c(Y_t - G_t) \frac{Y_t}{\mu_t} + \alpha \beta \mathbb{E}_t \left[\left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta-1} F_{t+1} \right],$$
$$J_t \equiv \xi_t \left(\frac{Y_t}{A_t} \right)^{1+\eta} + \alpha \beta \mathbb{E}_t \left[\left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta(1+\eta)} J_{t+1} \right].$$

Then arrive at $\left(\frac{1-\alpha(\Pi_t/\Pi)^{\theta-1}}{1-\alpha} \right)^{\frac{1+\theta\eta}{\theta-1}} = \frac{F_t}{J_t}$.

[Back to Aggregate Supply](#)

Derivation of Log-Linearized Equilibrium

We derive the first-order approximation of the AD equation (5.26)

$$\xi_t U_c(Y_t - G_t) = \beta(1 + i_t) E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(Y_{t+1} - G_{t+1}) \right\}. \quad (5.39)$$

Assume isoelastic utility of the form

$$U(C) = \frac{C^{1-\tilde{\sigma}^{-1}}}{1-\tilde{\sigma}^{-1}},$$

with $\tilde{\sigma} > 0$ and recall that $C_t = Y_t - G_t$.

Take logs of (5.39) to obtain

$$\ln \xi_t - \tilde{\sigma}^{-1} \ln C_t = \ln \beta + \ln(1 + i_t) + \ln E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(C_{t+1}) \right\}. \quad (5.40)$$

Note that in a log-linear approximation

$$\begin{aligned}
 \ln E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(C_{t+1}) \right\} &= \ln \left\{ \frac{1}{\bar{\Pi}} \bar{\xi} \bar{C}^{-\tilde{\sigma}^{-1}} \right\} + E_t \{ \ln \xi_{t+1} / \bar{\xi} + \\
 &\quad - \tilde{\sigma}^{-1} \ln C_{t+1} / \bar{C} - \ln \Pi_{t+1} / \bar{\Pi} \} \\
 &= \ln \bar{\xi} - \tilde{\sigma}^{-1} \ln \bar{C} - \pi + E_t \{ \hat{\xi}_{t+1} - \tilde{\sigma}^{-1} \hat{C}_{t+1} \\
 &\quad - (\pi_{t+1} - \pi) \}
 \end{aligned} \tag{5.41}$$

having used definitions $\pi_t \equiv \ln \Pi_t$, $\pi \equiv \ln \bar{\Pi}$ and where variables with hats denote log deviations of the respective variable from the steady state.

Combining (5.40) and (5.41), we can write

$$\hat{\xi}_t - \tilde{\sigma}^{-1} \hat{C}_t = \hat{i}_t + E_t \{ \hat{\xi}_{t+1} - \tilde{\sigma}^{-1} \hat{C}_{t+1} - (\pi_{t+1} - \pi) \} \quad (5.42)$$

having used definition $\hat{i}_t = \ln(1 + i_t)/\ln(1 + i)$ and noted that $\ln(1 + i) = -\ln \beta + \ln \Pi$. Consider now equilibrium condition $C_t = Y_t - G_t$, it can be written as

$$\frac{C}{Y} \frac{C_t - C}{C} = \frac{Y_t - Y}{Y} - \frac{G_t - G}{Y}$$

and therefore

$$\frac{C}{Y} \hat{C}_t = \hat{Y}_t - \hat{G}_t \quad (5.43)$$

noting that up to a first-order approximation $\hat{C}_t = \ln C_t/C = (C_t - C)/C$, $\hat{Y}_t = \ln Y_t/Y = (Y_t - Y)/Y$ and having used definition $\hat{G}_t = (G_t - G)/Y$.

Using (5.43) in (5.42), we obtain AD equation (5.35)

$$\hat{Y}_t = \hat{G}_t + E_t(\hat{Y}_{t+1} - \hat{G}_{t+1}) - \sigma \left(\hat{i}_t - E_t(\pi_{t+1} - \pi) + E_t\hat{\xi}_{t+1} - \hat{\xi}_t \right),$$

in which $\sigma = \tilde{\sigma}Y/C$.

We derive now the first-order approximation of the AS block represented by

$$\left(\frac{1 - \alpha \left(\frac{\Pi_t}{\Pi} \right)^{\theta-1}}{1 - \alpha} \right)^{\frac{1+\theta\eta}{\theta-1}} = \frac{F_t}{J_t}, \quad (5.44)$$

where F_t and J_t satisfy:

$$F_t = \xi_t (Y_t - G_t)^{-\frac{1}{\sigma}} \frac{Y_t}{\mu_t} + \alpha\beta E_t \left\{ F_{t+1} \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta-1} \right\}, \quad (5.45)$$

$$J_t = \xi_t \left(\frac{Y_t}{A_t} \right)^{(1+\eta)} + \alpha\beta E_t \left\{ J_{t+1} \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta(1+\eta)} \right\}. \quad (5.46)$$

Take logs of equation (5.44) to obtain

$$\log \left(\frac{1 - \alpha \left(\frac{\Pi_t}{\Pi} \right)^{\theta-1}}{1 - \alpha} \right) = -\frac{\theta - 1}{1 + \theta\eta} (\log J_t - \log F_t), \quad (5.47)$$

and also define

$$f_t \equiv \xi_t (Y_t - G_t)^{-\frac{1}{\theta}} Y_t \mu_t^{-1},$$

$$j_t \equiv \xi_t \left(\frac{Y_t}{A_t} \right)^{(1+\eta)}.$$

A first-order Taylor approximation of the left-hand side of (5.47) implies that

$$\log \left(1 - \frac{\alpha \left(\frac{\Pi_t}{\Pi} \right)^{\theta-1}}{1 - \alpha} \right) = -\frac{\alpha}{1 - \alpha} (\theta - 1) (\pi_t - \pi). \quad (5.48)$$

We must still derive a first-order approximation for $\log J_t$ and $\log F_t$ on the right-hand side of equation (5.47).

The definitions of J_t and F_t imply first-order expansions

$$\hat{F}_t = (1 - \alpha\beta)\hat{f}_t + \alpha\beta E_t\{\hat{F}_{t+1} + (\theta - 1)(\pi_{t+1} - \pi)\}, \quad (5.49)$$

$$\hat{J}_t = (1 - \alpha\beta)\hat{j}_t + \alpha\beta E_t\{\hat{J}_{t+1} + \theta(1 + \eta)(\pi_{t+1} - \pi)\}. \quad (5.50)$$

Moreover, note that

$$\frac{\alpha(1 + \theta\eta)}{1 - \alpha}(\pi_t - \pi) = (\hat{J}_t - \hat{F}_t), \quad (5.51)$$

using (5.47) and (5.48). Therefore,

$$\begin{aligned} (\pi_t - \pi) &= \frac{1 - \alpha}{\alpha(1 + \theta\eta)}((1 - \alpha\beta)(\hat{j}_t - \hat{f}_t) + \alpha\beta E_t\{\hat{J}_{t+1} - \hat{F}_{t+1} + (1 + \theta\eta)(\pi_{t+1} - \pi)\}) \\ &= \frac{1 - \alpha}{\alpha(1 + \theta\eta)}(1 - \alpha\beta)(\hat{j}_t - \hat{f}_t) + \beta E_t\{\pi_{t+1} - \pi\}. \end{aligned}$$

Note that

$$\hat{f}_t = \hat{\xi}_t - \sigma^{-1}(\hat{Y}_t - \hat{G}_t) + \hat{Y}_t - \hat{\mu}_t,$$

$$\hat{f}_t = \hat{\xi}_t + (1 + \eta)(\hat{Y}_t - \hat{A}_t).$$

We can finally derive the AS equation (5.33)

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi),$$

where

$$\kappa = \frac{(1 - \alpha)(1 - \alpha\beta)(\sigma^{-1} + \eta)}{\alpha(1 + \theta\eta)},$$

given the definition of the natural level of output

$$\hat{Y}_t^n = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t - \frac{1}{\sigma^{-1} + \eta} \hat{\mu}_t.$$

[Back to Log-Linearized Equilibrium](#)

8 Appendix: Derivations

9 Bibliography

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